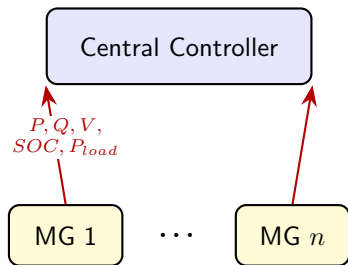
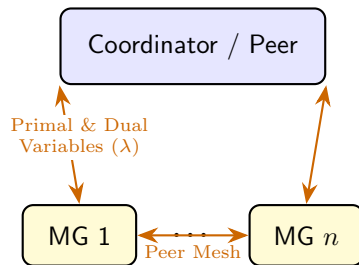


(a) Centralized OPF



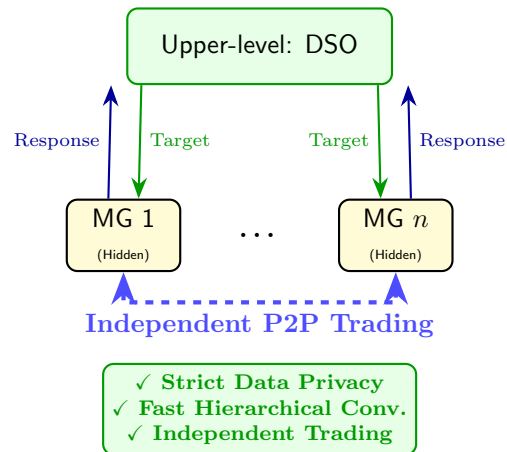
- × Severe Privacy Leak
- × Single-point Failure
- × Heavy Computation

(b) Distributed (ADMM)



- △ High Comm. Overhead
- △ Complex Coordination
- △ Slow Hierarchy Conv.

(c) Proposed Hierarchical ATC



- ✓ Strict Data Privacy
- ✓ Fast Hierarchical Conv.
- ✓ Independent Trading

(d) ATC Iterative Sequence Flow: Secure & Independent P2P Coordination

